

Running the study on Windows

For a second researcher setting up their own machine.

The flow is the same on Windows as on a Mac: **open the project in Unity, press Build And Run, then paste one command into a terminal**. Nothing about the study itself differs. Only the shape of that one command changes, and there is a `study.cmd` in this repository that makes even that identical.

One-time setup: about an hour, most of it Unity downloading in the background. After that, starting a session takes a minute.

What you need

A Windows laptop	Where you build from and where the data is saved.
The Quest headset	In developer mode, with a USB-C cable that carries data.
Wi-Fi	The laptop and the headset on the same network.

A charge-only USB cable will never show the headset and looks identical to a data one. If the headset is invisible later, suspect the cable first.

Part 1 — One-time setup

Step 1 — Unity Hub and Unity 6000.3.19f1

1. Get **Unity Hub** from <https://unity.com/download> and install it.
2. Sign in (a free personal licence is fine) and activate it.
3. Hub → **Installs** → **Install Editor** → **Archive** tab → the *"download archive"* link.
4. Find **Unity 6000.3.19f1** and click its **Unity Hub** button.

The version has to match exactly. A different version silently re-imports and upgrades the project, and the build stops matching the one the other researcher is running. Two researchers running different builds is not a study.

5. On the modules screen tick **all three**:
 - **Android Build Support**
 - **OpenJDK** (nested under it)
 - **Android SDK & NDK Tools** (nested under it)

This is what makes a headset build possible. Adding it later means another long download.

6. Install. Several gigabytes, so give it time.

Step 2 — Node.js

<https://nodejs.org> → the green **LTS** button → Next through the installer. Nothing needs changing.

Step 3 — Git

<https://git-scm.com/download/win> → Next through it.

Step 4 — Get the project

Open **Command Prompt** and run these two lines:

```
cd %USERPROFILE%\Desktop
git clone -b Visualisation-DreamCodeVR-feedback-loop https://github.com/abyyworld/Real-Time-Immersive-Speech-Programming.git dreamcodevr
```

That last word matters: it puts everything in a short folder, `Desktop\dreamcodevr`, which every path below assumes.

If that URL asks for credentials you do not have, ask for access, or for the supervisor's copy of the repository instead. Do not download a ZIP of the default branch — the study lives on the branch named above.

Step 5 — Headset developer mode

1. Pair the headset in the **Meta Horizon** phone app.
2. Headset → **Settings** → **System** → **Developer** → turn developer mode and **USB Connection Dialog** on.
3. Plug the headset into the laptop.
4. **Put the headset on.** There is an *"Allow USB debugging?"* prompt waiting inside it — tick **Always allow**, tap **Allow**. You cannot see or accept that prompt from the laptop. This is the single most common reason the headset appears to be invisible.

Part 2 — Building onto the headset

Step 6 — Open the project

Unity Hub → **Open** → select `Desktop\dreamcodevr\Unity`.

Select the **Unity** folder inside, not the outer `dreamcodevr` folder.

The first open takes **15–40 minutes** while it compiles the project. It will look frozen. It is not. Let it finish.

Step 7 — Switch the platform to Android

This is the step that catches everyone.

A fresh clone opens set to **Windows**, because that is the machine it is on. The active platform is stored in **Library/**, which is deliberately not in the repository, so it cannot travel with the project.

If you press Build And Run now you get a Windows program on your Desktop in about twenty seconds, **no error**, and nothing on the headset. It looks like it worked.

So, before building:

1. **File → Build Profiles**
2. Select **Android**
3. Click **Switch Platform**

10–30 minutes the first time, because it re-compresses every texture for Android. Once only.

You will know it worked because **Android** now has the Unity logo beside it in that window.

Step 8 — Build And Run

Headset plugged in and awake:

File → Build Profiles → Build And Run

- The **first** build takes **30–40 minutes**. That is normal — it compiles the engine to native code. Later builds are a few minutes.
- **Do not unplug the headset**. If it sleeps and drops off USB near the end, the build itself is fine but installing fails. Plug it back in and press Build And Run again; the second attempt is quick.

When it finishes, the app launches in the headset by itself.

Afterwards you find it under **Apps → Unknown Sources → DreamCodeVR**. It does not appear in the normal app list — that is where sideloaded apps live, and is expected.

Part 3 — Running a session

The whole routine, every time.

Step 9 — Start the study server

Open **Command Prompt** and paste:

```
%USERPROFILE%\Desktop\dreamcodevr\study
```

This is the same command the other researcher runs, pointed at your Desktop. Windows resolves it to **study.cmd**; a Mac resolves it to the **study** script beside it. Same instruction, either machine.

It asks which mode you want, then starts everything.

The first time, it spends a minute or two installing what the server needs. Once only.

Windows will ask for firewall permission — `"Allow Node.js to communicate on these networks"`. Tick **Private networks** and **Allow access**. If you click Cancel the headset cannot reach the laptop; close the window, run the command again, and allow it that time.

Leave that window open for the whole session. Closing it stops everything. Minimising is fine.

Step 10 — Open the panel

The window prints an address like:

```
Panel: http://192.168.1.42:8181
```

Open it in your browser. That is where you run the session from.

Step 11 — Put the headset on

Open **DreamCodeVR** (Apps → Unknown Sources). It finds the laptop by itself. There is no address to type anywhere, and it works on whatever Wi-Fi you are both on that day.

Do a dry run before your first participant

Do not let the first time you run this be with someone sitting in the room.

With the **USB cable plugged in**, go through steps 9–11 and check that:

- The panel opens in the browser.
- The headset app connects, rather than sitting on "Looking for server".
- The panel shows the microphone picking up when you speak.

Keep the cable in for that test. With the cable in, laptop and headset can always reach each other regardless of the Wi-Fi, so it separates *is the software working* from *is the network cooperating*.

Where the data goes

```
Desktop\dreamcodevr\Logs\
```

One CSV per participant — `P01.csv`, `P02.csv`. Everything about that person is in their one file.

Back these up after every session. They are kept out of version control on purpose: participant data is never uploaded. That also means nothing is protecting them but you. Copy them to OneDrive or a memory stick the same day.

Agree participant numbers with the other researcher before you start, so you do not both create a `P02`. The simplest split is odd numbers for one of you and even for the other.

Troubleshooting

Build And Run finished in 20 seconds and nothing is on the headset You are still on the Windows platform. Go back to Step 7.

The headset sits on "Looking for server" Usually the Wi-Fi blocking devices from seeing each other, which is normal on university networks. Plug the USB cable in and run the `study` command again — it tunnels the connection down the cable and the Wi-Fi stops mattering.

Otherwise check both are on the same network, not one on eduroam and one on a guest network.

It printed the wrong address, or 127.0.0.1 Laptops with a VPN or virtual machines sometimes have the wrong adapter picked. Find the real one with `ipconfig` (it starts `192.168.` or `10.`), then:

```
set STUDY_LAN_IP=192.168.1.42
%USERPROFILE%\Desktop\dreamcodevr\study
```

using your actual address.

Unity cannot find the headset Charge-only cable, or the *"Allow USB debugging?"* prompt was never accepted. Put the headset on and look for it.

The app is not hearing the participant Check the headset's microphone is not muted — Quick Settings inside the headset. This is a headset setting, nothing to do with the study software, and easy to knock on by accident.

'study' is not recognized as an internal or external command Either the path is wrong, or you are in PowerShell rather than Command Prompt. PowerShell needs `& "%USERPROFILE%\Desktop\dreamcodevr\study"` with the ampersand and quotes. Command Prompt is simpler.

If something goes wrong

Screenshot the terminal window with the error in it and send it on. The text in that window is what makes a problem fixable; "it did not work" on its own is very hard to act on.

Never delete anything in `Logs\`.